

Gopikrishnan Chirappurathu Remesan

Résumé

October 15, 2021

Curriculum Vitae

Given names : **Gopikrishnan** Last name : **Chirappurathu Remesan**
Citizenship : Indian Date of birth : 26th May, 1992
Languages : Malayalam (first language), English (fluent)
Marital status : Married
Current address : Dept. of Mathematics, Indian Institute of Technology Bombay, Powai,
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ORCID: [0000-0002-4507-4463](https://orcid.org/0000-0002-4507-4463) Scopus ID: [57219814720](https://scopus.com/authid/detail.uri?authorid=57219814720) MR Author ID: [1405273](https://www.mr-author.org/author/1405273)

Current position

July-Dec 2021 (on going) **Post doctoral fellow**, *Dept. of Mathematics, Indian Institute of Technology Bombay, Maharashtra, India.*
Topic: Numerical analysis of finite element methods for phase field crystal equations, a priori and a posteriori error estimates for finite element methods for fourth order semilinear partial differential equations. **Mentor:** Prof Neela Nataraj.
Jan 2022 - Feb 2023 **Post doctoral fellow**, *School of Mathematics, The University of Milan, Italy.*
Topic: Development, analysis, and numerical testing of a posteriori estimates, near best approximation in a given, possibly nonconforming discrete solution space, best error localizations in the adaptive solution of a partial differential equation with arbitrary variational data. **Mentor:** Prof Andreas Veiser.

Education

Integrated BS – MS Dual Degree

2010 - 2012 **Bachelor of Science (Biology, Chemistry, Physics, and Mathematics)**, *Indian Institute of Science Education and Research, Thiruvananthapuram, India*, Cumulative grade point average - 8.65/10 up to 4th semester.
2012 - 2015 **Master of Science (Mathematics)**, *Indian Institute of Science Education and Research, Thiruvananthapuram, India*, Cumulative grade point average - 8.58/10.

Doctor of Philosophy

- 2016 July - **Doctor of Philosophy (Mathematics)**, *IITB – Monash Research Academy*,
2021 June Joint doctoral programme by Indian Institute of Technology Bombay, India
and Monash University, Australia on the topic “**Modelling and numerical
analysis of complex tumour growth problems**”,
Thesis defended on : June 18, 2021
PhD awarded on : August 7, 2021.
See a [Three – Minute Talk](#) on my PhD thesis.
See a [One – Minute Animation](#) on my PhD thesis.
- Thesis advisors** IIT Bombay: Prof Neela Nataraj
Monash University: Prof Jerome Droniou, A/Prof Jennifer Flegg

Honours and awards

- 2010 - 2015 **INSPIRE** (Innovation in Science Pursuit for Inspirational Research) Fellow-
ship (Department of Science and Technology, Government of India), Registra-
tion Number - DST/INSPIRE-SHE/IISER-T/2008
- 2012 - 2015 **Institute Silver Medal**, For best academic performance and highest cumula-
tive grade point average, School of Mathematics, IISER Thiruvananthapuram,
Kerala, India.
- 2020 August **First prize**, Three Minute Thesis Talk, IITB - Monash Research Academy.
- 2021 July **Prof. Prabhulal Bhatnagar Memorial Prize** (2020-2021) for being the
most outstanding of all the students who completed the requirements for the
degree of Doctor of Philosophy in Mathematics.

National level examinations

**Council of Scientific and Industrial Research - Junior Research Fel-
lowship**, June 2015 (AIR 89/768), December 2015 (AIR 43/934), June 2016
(AIR 123/1805).

Graduate Aptitude Test in Engineering – Mathematics, January 2016
(AIR 104/6305).

Experience

- June 2015 - **Visiting Lecturer**, *Bishop Chulapparambil Memorial College Kottayam*, Ker-
July 2016 ala, India.
- 2016–2020 Teaching assistantships at IIT Bombay
- Calculus (IIT Bombay, B.Tech. first year, 2018)
 - Numerical analysis (IIT Bombay, B.Tech. second year, 2018)
 - Numerical analysis (IIT Bombay, M.Sc. first year, 2019)
 - Numerical analysis (IIT Bombay, M.Sc. first year, 2020)
 - Programming Lab (Fortran) (IIT Bombay, M.Sc. first year, 2021)

April 2021 - **Research associate**, *Dept. of Mathematics, Indian Institute of Technology*
June 2021 *Bombay, Maharashtra, India.*

Research interest

Main research interest are mathematical modelling of different physical problems (with focus on problems from life sciences), design and implementation of numerical schemes, and theoretical and numerical analysis. This entails:

- ✦ Derivation of mathematical models using multiphase fluid flow and mixture theory, linear and nonlinear elasticity.
- ✦ Design of numerical schemes using appropriate combinations of finite volume, finite element, and finite difference methods.
- ✦ Rigorous theoretical and numerical analysis of physical models with hyperbolic, elliptic, and parabolic partial differential equations.

Another research interest is numerical analysis (and implementation) of finite element methods for elliptic partial differential equations such as biharmonic, Navier-Stokes, and von Kármán equations.

Publications

- [1] G. C. Remesan. Strong bounded variation estimates for the multi-dimensional finite volume approximation of scalar conservation laws. *ESAIM:M2AN*, 55(4), 2021. URL <https://doi.org/10.1051/m2an/2021027>.
- [2] J. Droniou, N. Nataraj, and G. C. Remesan. Convergence analysis of a numerical scheme for a tumour growth model. *IMA Journal of Numerical Analysis*, 2021. URL <https://doi.org/10.1093/imanum/drab016>.
- [3] J. Droniou, J. Flegg, and G. C. Remesan. Numerical solution of a two dimensional tumour growth model with moving boundary. *Journal of Scientific Computing*, 85(20), 2020. URL <https://doi.org/10.1007/s10915-020-01326-6>.
- [4] G. C. Remesan. Numerical solution of the two-phase tumour growth model with moving boundary. In B. Lamichane, T. Tran, and J. Bunder, editors, *Proceedings of the 18th Biennial Computational Techniques and Applications Conference , CTAC-2018*, volume 60 of *ANZIAM J.*, pages C1–C15, 2019. URL <https://doi.org/10.21914/anziamj.v60i0.13936>.
- [5] H. M. Byrne, J. A. Flegg, and G. C. Remesan. Two phase model for compressive stress induced on a surrounding medium by an expanding tumour. *[In preparation]*.

Technical skill

Programming C, C++, Python, Fortran
Language

Softwares Matlab, Mathematica, L^AT_EX, Paraview

Teaching interests

Ordinary and partial differential equations	Mathematical analysis
Numerical analysis	Multivariable calculus

In numerical analysis, special interests are **finite difference**, **finite volume**, and **finite element methods**. In differential equations, **variational theory based on Sobolev spaces** along with the classical theory of differential equations are topics of interest.

Contributed talks

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| November 2017 | Popular talk on mathematics on the topic ‘ <i>Buffon’s needle problem and what is so harmonic</i> ’, IIT Bombay, India. |
| October 2018 | Postgraduate student talk on the topic ‘ <i>Mathematics and medicine: the common ‘M’’</i> ’, Monash University, Australia. |
| November 2018 | As a part of CTAC 2018 conference on the topic - ‘ <i>Numerical solution of the two-phase tumour growth model with moving boundary (1-spatial dimensional study)</i> ’, Newcastle, Australia. |
| April 2019 | Talk as a part of MCB lecture series on the topic ‘ <i>Numerical solution of a two-phase tumour growth model with moving boundary (2 spatial dimensional study)</i> ’, University of Melbourne, Australia. |
| June 2019 | Talk on ‘ <i>Numerical solution of a two-phase tumour growth model in two spatial dimensions</i> ’, MAFELAP 2019, Brunel University, London. |
| June 2019 | Informal talk on ‘ <i>Numerical solution of a two-phase tumour growth models</i> ’, School of Mathematics, University of Oxford, London. |
| February 2020 | Talk on ‘ <i>Numerical solutions of a two dimensional tumour growth model</i> ’, Annual conference of ANZIAM 2020, New Castle, Australia. |
| February 2020 | Talk on ‘ <i>Convergence analysis of a two-phase tumour growth model</i> ’, MWN-DEA 2020, Monash University, Australia. |
| January 2021 | Talk on ‘ <i>Numerical solution of a tumour growth model in two spatial dimensions</i> ’, Annual conference of AustMS, Australia. |
| February 2021 | Talk on ‘ <i>Two-phase model for compressive stress induced on a surrounding hyperelastic medium by an expanding tumour</i> ’, Annual conference of ANZIAM 2021, Australia. |

Talks as a resource person

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| August 2015 | A talk on the topic ‘ <i>Measure Theory - When observed from grounds</i> ’ at St. Stephans College Kottayam, Kerala, India. |
| December 2019 | Workshop on ‘ <i>Data visualisation using Matlab</i> ’, IIT Bombay (organised by Research Scholars Forum), India. |
| December 2019 | Talk on ‘ <i>Overdetermined systems of linear equations</i> ’, B. C. M. College, Kottayam, India. |
| December 2020 | Workshop on ‘ <i>Scientific computing and visualization with Matlab and Paraview</i> ’, IIT Bombay (organised by Post Graduate Academic Council), India. |

July 2021 A talk on the topic '*Finite element method in one dimension*' at St. Stephens College Kottayam, Kerala, India.

Projects

Bachelor of Science

Title **Electronic structure, lattice energies and Born exponents for alkali halides from first principles**, AIP Advances, 2(1), 2012, URL <https://doi.org/10.1063/1.3684608>.

Advisor Prof Ayan Datta, Professor, Indian Association for the Cultivation of Sciences, India

Master of Science

Title **A theoretical and numerical study of stochastic delay integro differential equations**

Advisor Prof M. P. Rajan, Professor, School of Mathematics, IISER Thiruvananthapuram, India.

Master of Science (for minor degree in physics)

Title **Universal behavior of quantum discord as a function of measurement strength**

Advisor Prof Anil Shaji, Professor, School of Physics, IISER Thiruvananthapuram, India.

Participated workshops, seminars, and conferences

December 2013 Workshop - Winter school in probability, Indian Statistical Institute Kolkata, India.

August 2015 Refresher course for College teachers in Kerala (Topology), Kerala School of Mathematics, Kozhikode, Kerala, India.

June 2017 Conference: Recent advances in PDE: theory, computations and applications, IIT Bombay, India.

March 2018 Workshop: New directions in PDE constrained optimisation, IIT Bombay, India.

November 2018 Conference: Computational Techniques and Applications Conference, Newcastle, Australia.

February 2019 Workshop: Python for mathematics, Monash University, Australia.

June 2019 Conference: Mathematics of Finite Elements and Applications - 2019, Brunel University, London.

August 2019 Workshop: Hyperbolic conservation laws - Theory and Numerics, TIFR-CAM, Bangalore, India.

February 2020 Conference: Australian and New Zealand Industrial and Applied Mathematics Conference, New Castle, Sydney, Australia.

- February 2020 Workshop: Monash Workshop on Numerical Differential Equations and Applications, Monash University, Australia.
- January 2021 Conference: Annual conference of Australian Mathematical Society.
- February 2021 Conference: Annual conference of Australian and New Zealand Industrial and Applied Mathematics.

References

1. Prof Neela Nataraj, Institute Chair Professor, Dean (Faculty Affairs), Dept. of Mathematics, Indian Institute of Technology Bombay, Mumbai, Maharashtra, India.
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2. Prof Jerome Droniou, Professor, School of Mathematics, Monash University, Australia.
☎ : +61 3 9905 4489, ✉ : jerome.droniou@monash.edu
3. A/Prof Jennifer Flegg, Associate professor, School of Mathematics and Statistics, University of Melbourne, Australia.
☎ : +61 3 8344 7523, ✉ : jennifer.flegg@unimelb.edu.au